

PRODUCT ANALYTICS

CONJOINT ANALYSIS PRACTICE

This example will show you to analyze how consumers make tradeoffs among different attributes of a new product. Dell wants to launch a new laptop with 4GB RAM which Lenovo and HP already have in the market. Following are the attributes and levels were finalized: five attributes (e.g., 1: brands, 2: processor speed, 3: screen sizes, and 4: RAM and 5: Price). Following are the level of each attributes.

Attributes:	Brand:	: Lenovo, HP, and Dell
	Processor speed	: Dual core, Quad core
	Screen Size	: 15, 17, and 21 inches.
	RAM	: 2 GB, 4 GB, 8 GB
	Price	: Rs. 40000, Rs 45000. Rs 55000,

Choosing each level of the five attributes, we can make total $3 \times 2 \times 3 \times 3 \times 3 = 162$ combinations. For each combination, prepare a card similar to the following:

Card 1	
Brand:	Lenovo
Processor:	Dual core
Screen:	15 inch
RAM:	8 GB
Price	Rs 55000

Using the same pattern as in the following example, we need only the 16 combinations for the estimation of the part-worth utilities. Further, prepare *any* three more cards for the validation.

16 Combinations for the estimation

Card No.	Brand	Processor	Screen	RAM	Price	Your Preference Rating
1	Dell	Quad core	21 inch	8 GB	Rs 45000	
2	HP	Dual core	17 inch	8 GB	Rs. 40000	
3	Lenovo	Quad core	21 inch	4 GB	Rs. 40000	
4	HP	Quad core	15 inch	2 GB	Rs 45000	
5	Lenovo	Quad core	15 inch	8 GB	Rs. 40000	
6	Lenovo	Dual core	15 inch	8 GB	Rs 45000	
7	HP	Quad core	15 inch	4 GB	Rs 55000	
8	Lenovo	Dual core	17 inch	4 GB	Rs 45000	
9	Dell	Quad core	17 inch	8 GB	Rs 55000	
10	Dell	Dual core	15 inch	2 GB	Rs. 40000	
11	Lenovo	Dual core	15 inch	8 GB	Rs 55000	
12	HP	Dual core	21 inch	8 GB	Rs 55000	
13	Lenovo	Quad core	15 inch	8 GB	Rs 55000	
14	Dell	Dual core	15 inch	4 GB	Rs 55000	
15	Lenovo	Quad core	17 inch	2 GB	Rs 55000	
16	Lenovo	Dual core	21 inch	2 GB	Rs 55000	

3 Combinations for the validation

Card No.	Brand	Processor	Screen	RAM	Price	Your Preference Ratings
1	Lenovo	Quad core	21 inch	8 GB	Rs. 40000	
2	Dell	Dual core	17 inch	2 GB	Rs 55000	
3	Lenovo	Dual core	15 inch	4 GB	Rs. 40000	

1. You assign a rating score of your preference of each of the product profile. The ratings should range between 0 and 10 on **each** card (10 extremely preferred). Scores need not be integer. Thus, you would obtain 19 scores such as 3.4, 5.6, 2.7, ... , 9.5.
2. Using the rating scores of the first 16 combinations, run a linear regression with dummy variables for the estimation of the part-worth utilities. practice.sav file contains:

$$\text{Score} = \alpha + \beta_{HP}HP + \beta_{DL}DL + \beta_{P2.1}P_{quad} + \beta_{S17}S17 + \beta_{S21}S21 + \beta_{R1}R4 + \beta_{R2}R8 + \beta_{P45}P45 + \beta_{P50}P55 + \varepsilon$$

where *HP*, *DL*, *Pquad*, *S17*, *S21*, *R4*, *R8*, *P55*, and *P45* are dummy variables.

α represents the mean part-worth utility of the base model of the laptop: with brand Lenovo, processor Dual core, screen 15 inch, 2GB RAM and Rs. 40000 price.

β_{HP} = The mean difference of part-worth utilities between brand HP and Lenovo

β_{DL} = The mean difference of part-worth utilities between brand Dell and Lenovo

β_{Pquad} = The mean difference of part-worth utilities between processor Quad core and Dual core

β_{S17} = The mean difference of part-worth utilities between screen 17 inch and 15 inch

β_{S21} = The mean difference of part-worth utilities between screen 21 inch and 15 inch

β_{R4} = The mean difference of part-worth utilities between RAM 4 GB and 2 GB

β_{R6} = The mean difference of part-worth utilities between RAM 8 GB and 2 GB

β_{P45} = The mean difference of part-worth utilities between price Rs. 45K and Rs. 40K

β_{P55} = The mean difference of part-worth utilities between price Rs. 55K and Rs. 40K.

- a) Estimate the part-worths.
- b) Compute the utilities of the last 3 combinations (validation) using the estimated part-worths.
- c) Calculate importance of the attributes.